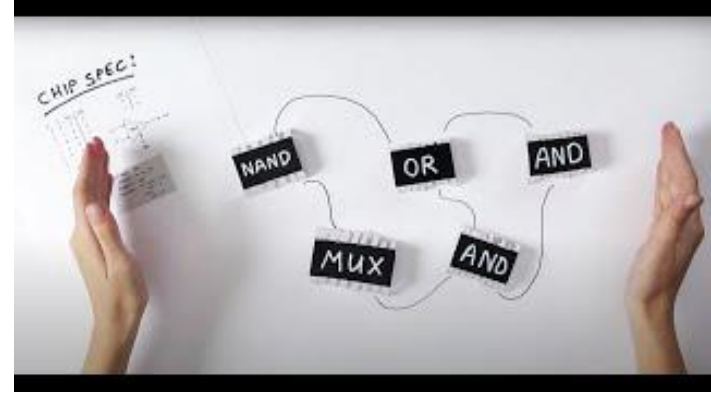




CSc 360

Operating Systems

Processes

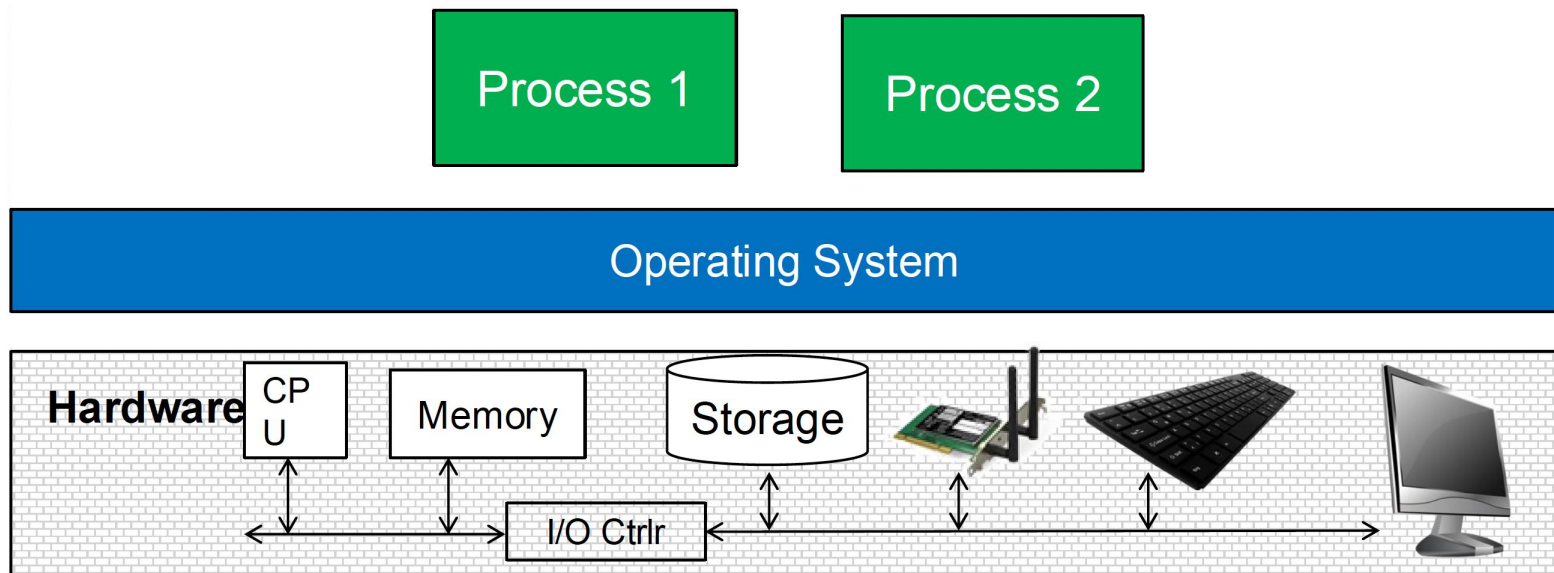
Wenjun Yang

Summer 2026

Recall: Topic Breakdown (Course Objective)

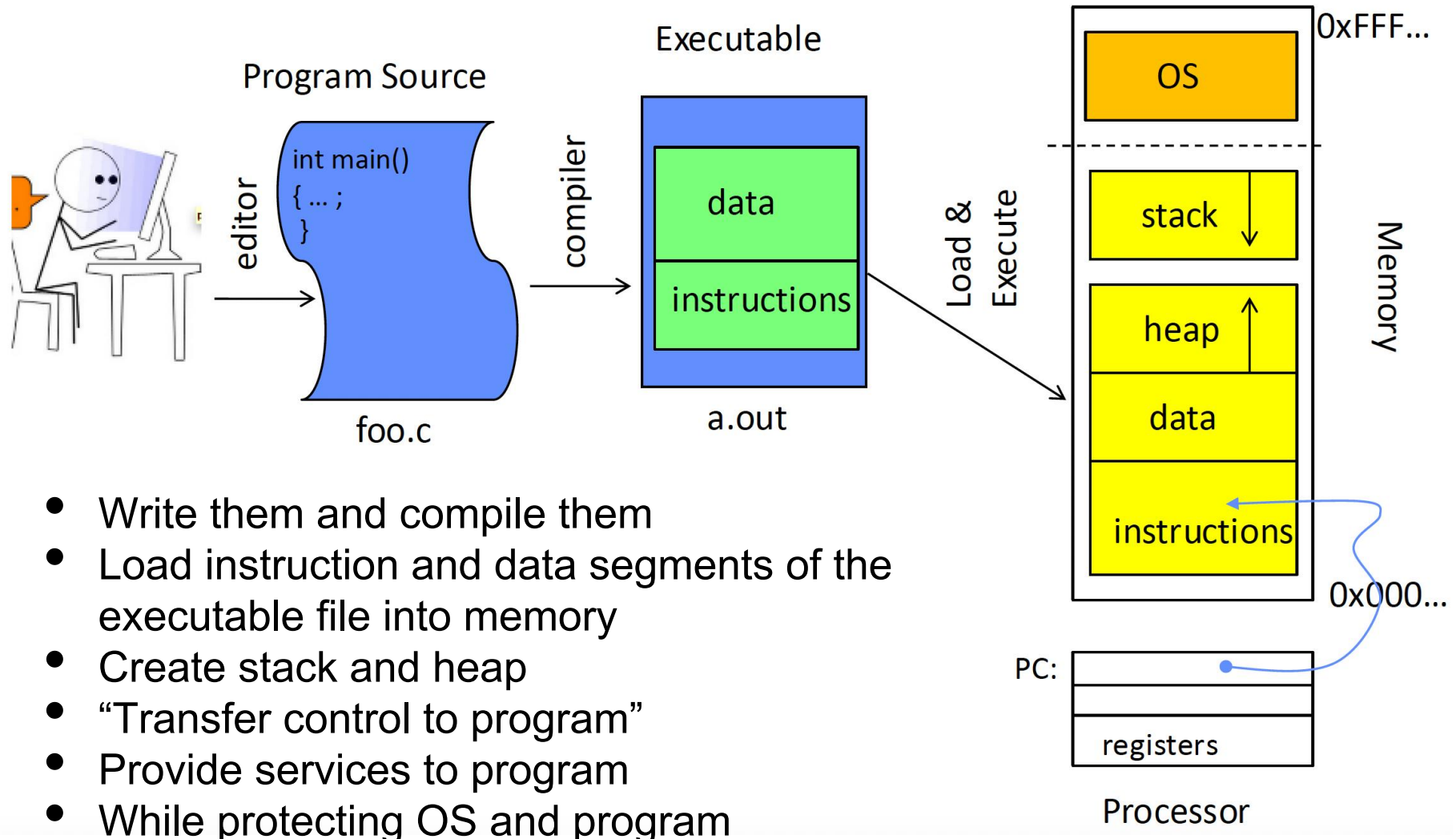
- virtualizing the CPU (**process**):
process, thread, scheduling, synch
- virtualizing memory (**memory**):
memory management, virtual memory
- **storage**: file systems, I/O systems

Process: an active instance of a program



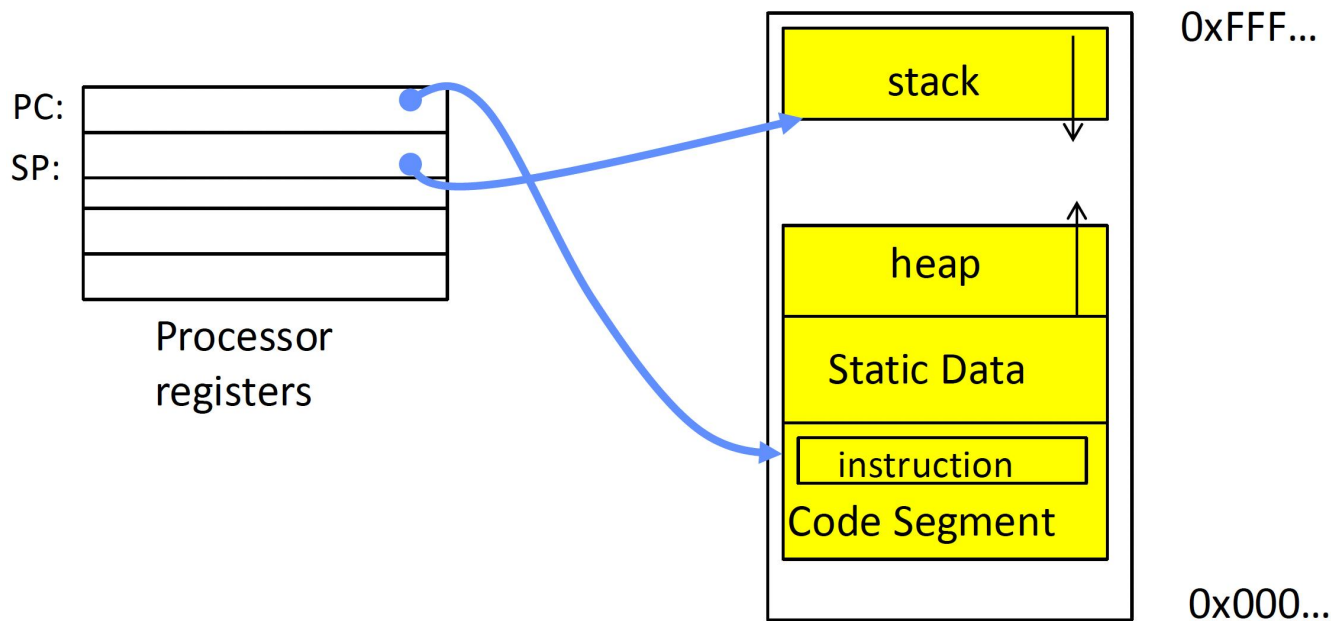
- OS virtualizes hardware to processes (applications) to provide the illusion that each process uses its own machine; also provides illusion of infinite memory and processor

OS Bottom Line: Run Programs



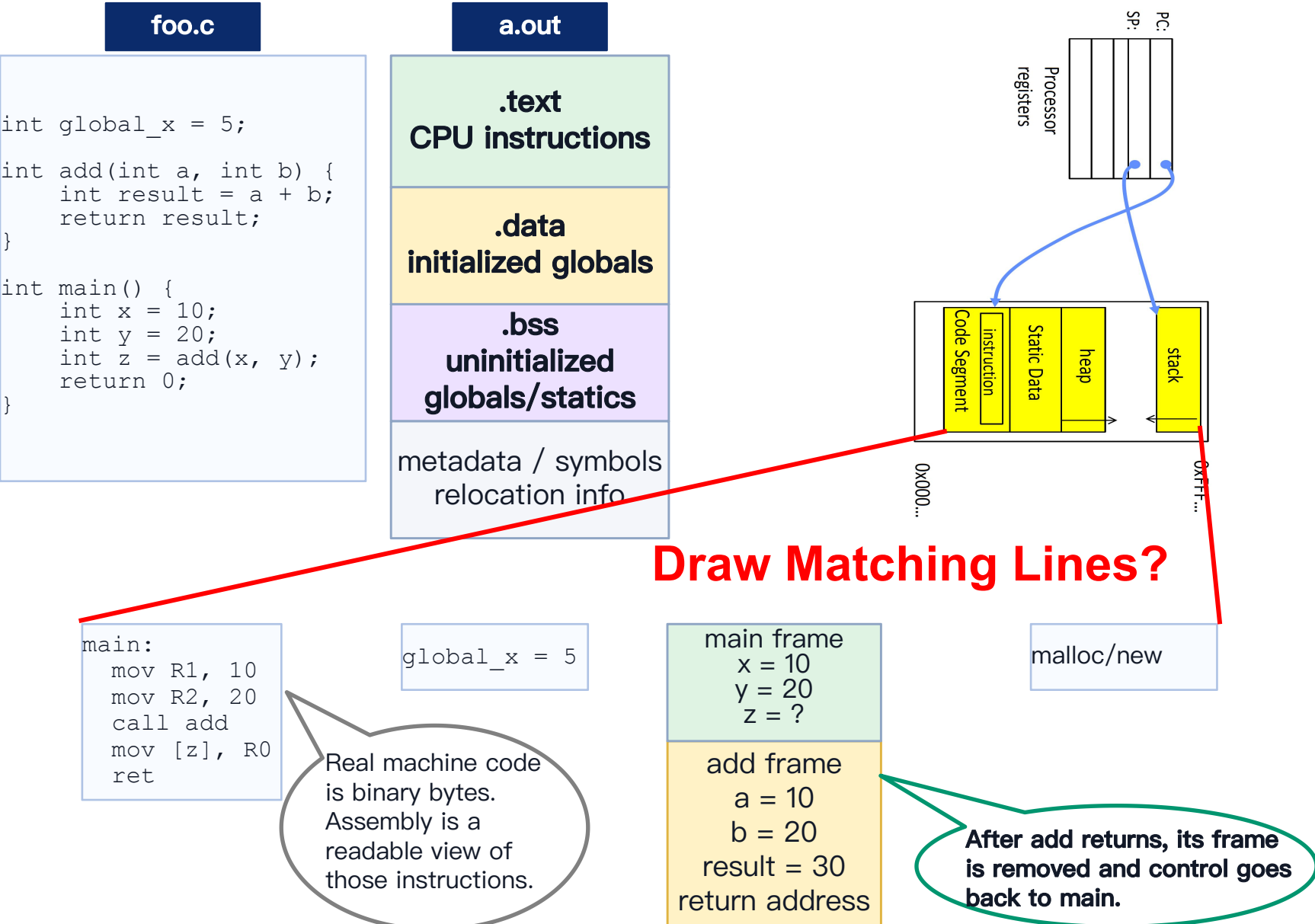
- Write them and compile them
- Load instruction and data segments of the executable file into memory
- Create stack and heap
- “Transfer control to program”
- Provide services to program
- While protecting OS and program

Address Space



- What's in the code segment? Static data segment?
- What's in the stack segment?
 - How is it allocated? How big is it?
- What's in the heap segment?
 - How is it allocated? How big?

OS Creates a Process

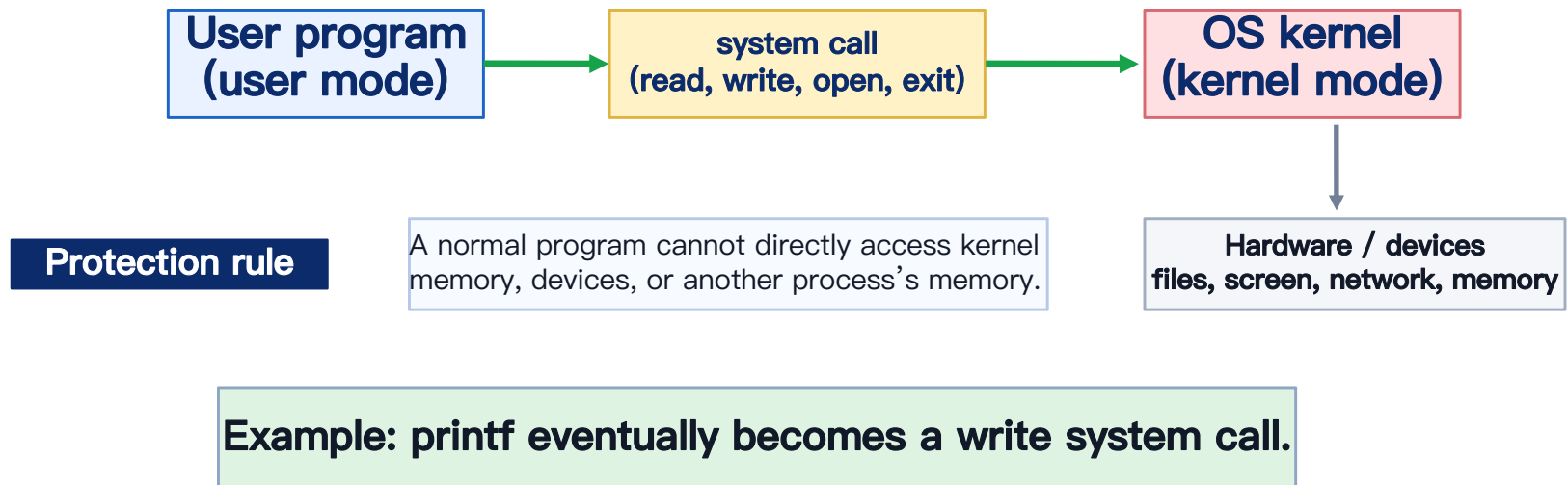


OS Jobs

- **Create Process**

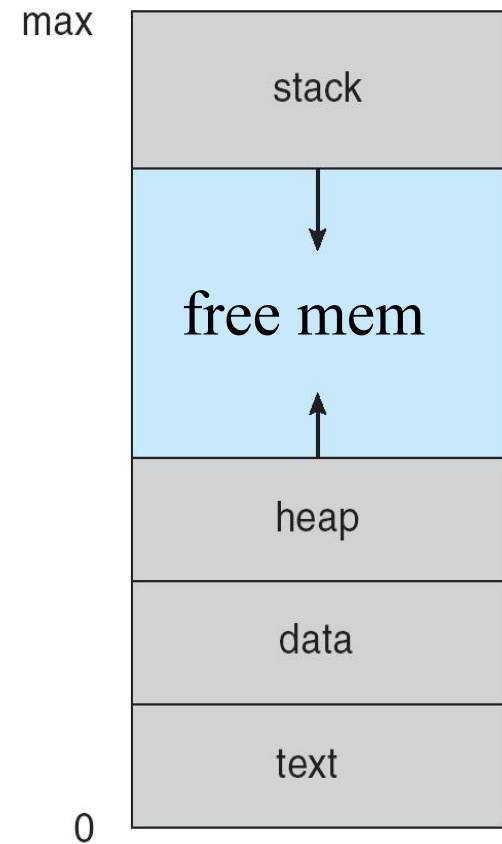
- Load .text, .data, .bss into memory
- Create stack and heap
- Initialize registers
- Set PC to entry point
- Transfer control to user code

- **Privileged kernel services**



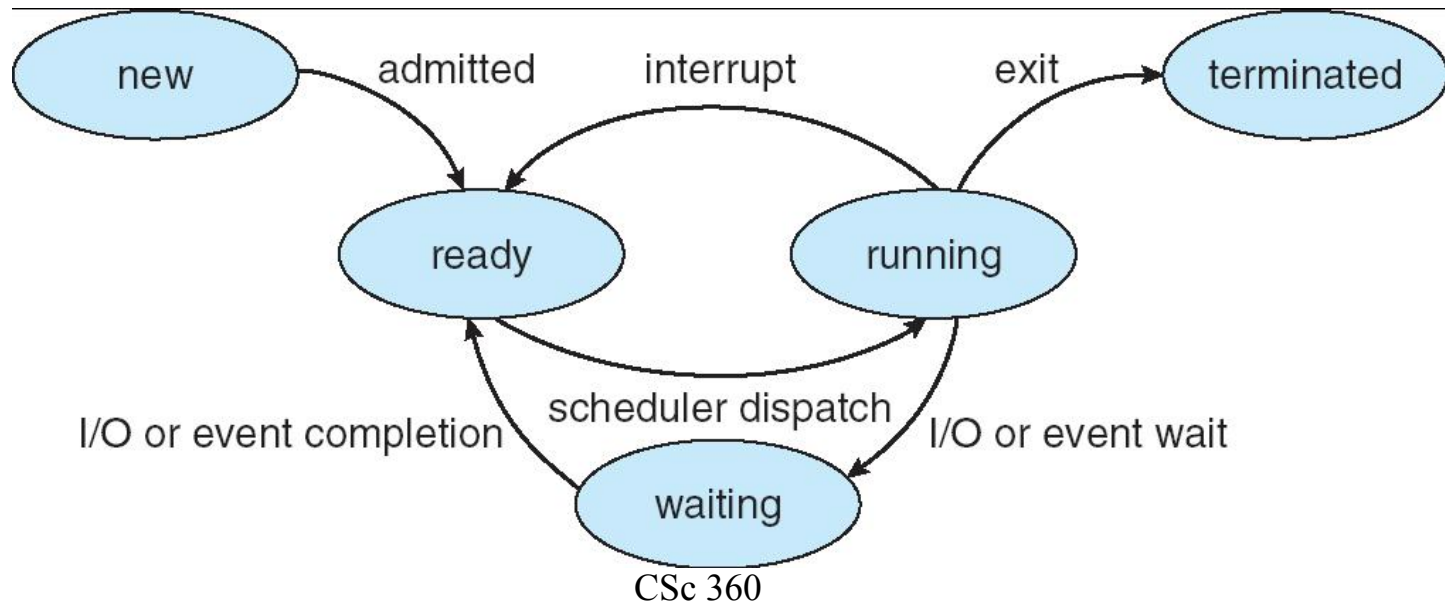
Processes

- Process: a program in execution
- Program: passive entity
 - static binary file on storage
 - e.g., `gcc -o hello hello.c`
- Process: active entity; resource allocated!
 - `./hello`
 - text (code); data (static), stack, heap
 - process control block (PCB)



Process states

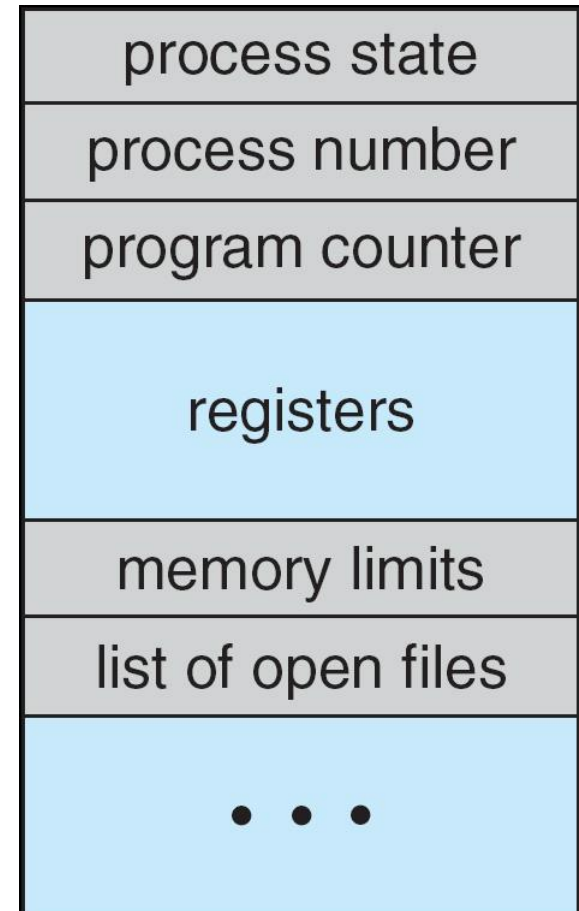
- E.g., one CPU (core)
 - one running process at any time
 - maybe many ready/waiting processes



why interrupt?

Process control blocks

- PCB: keep track processes
 - state: ready/running, etc
 - CPU
 - PC, registers, priority, etc
 - memory
 - memory control information
 - I/O
 - e.g., list of opened files
 - accounting

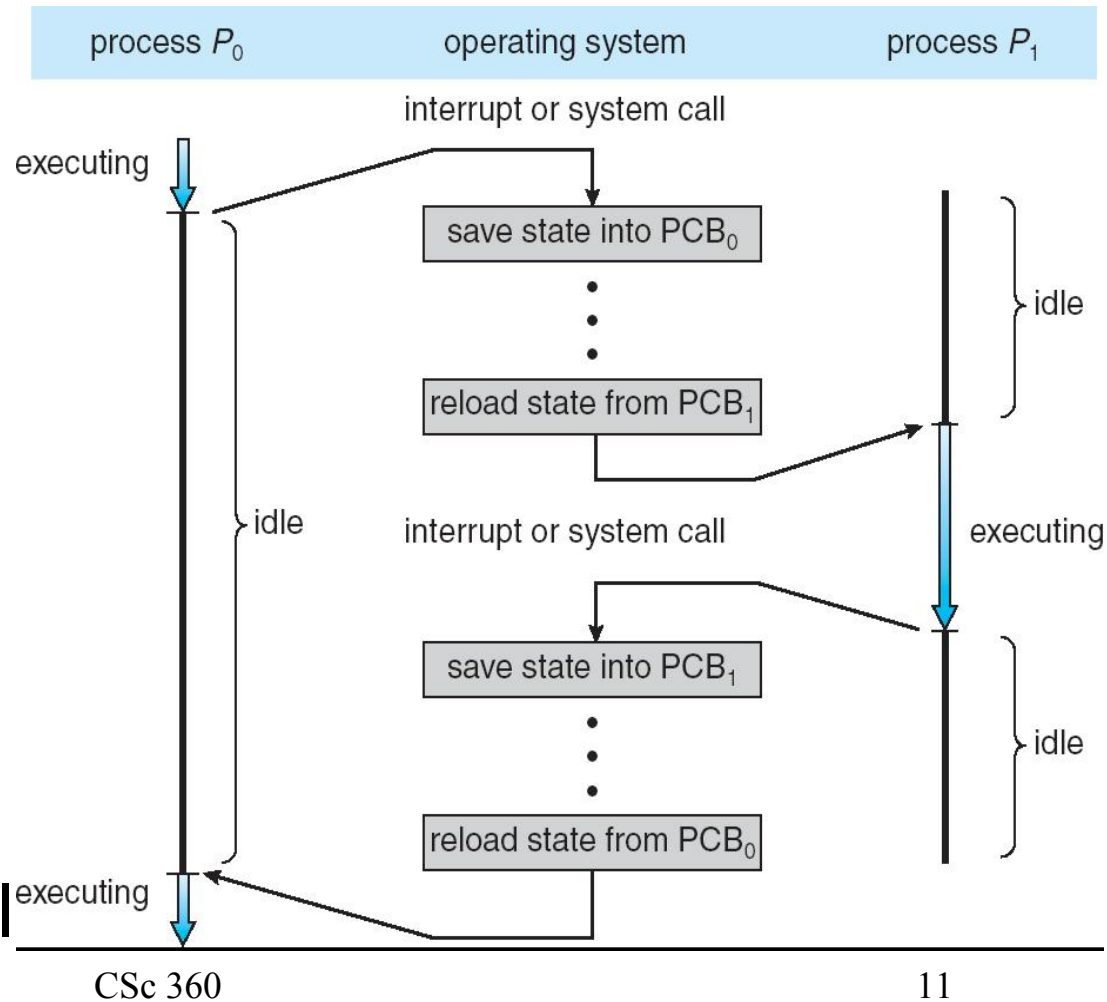


Context switching

- Context switch
 - save states
 - restore states

- When

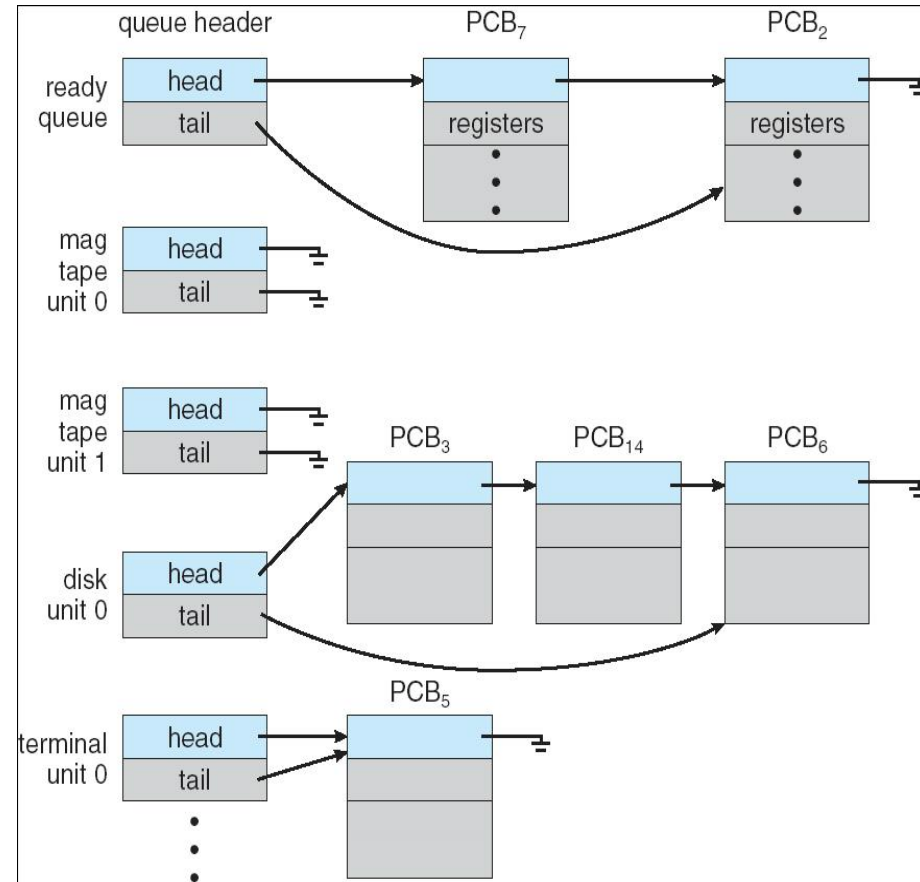
- timer
- I/O, memory
- trap
- waiting sys call



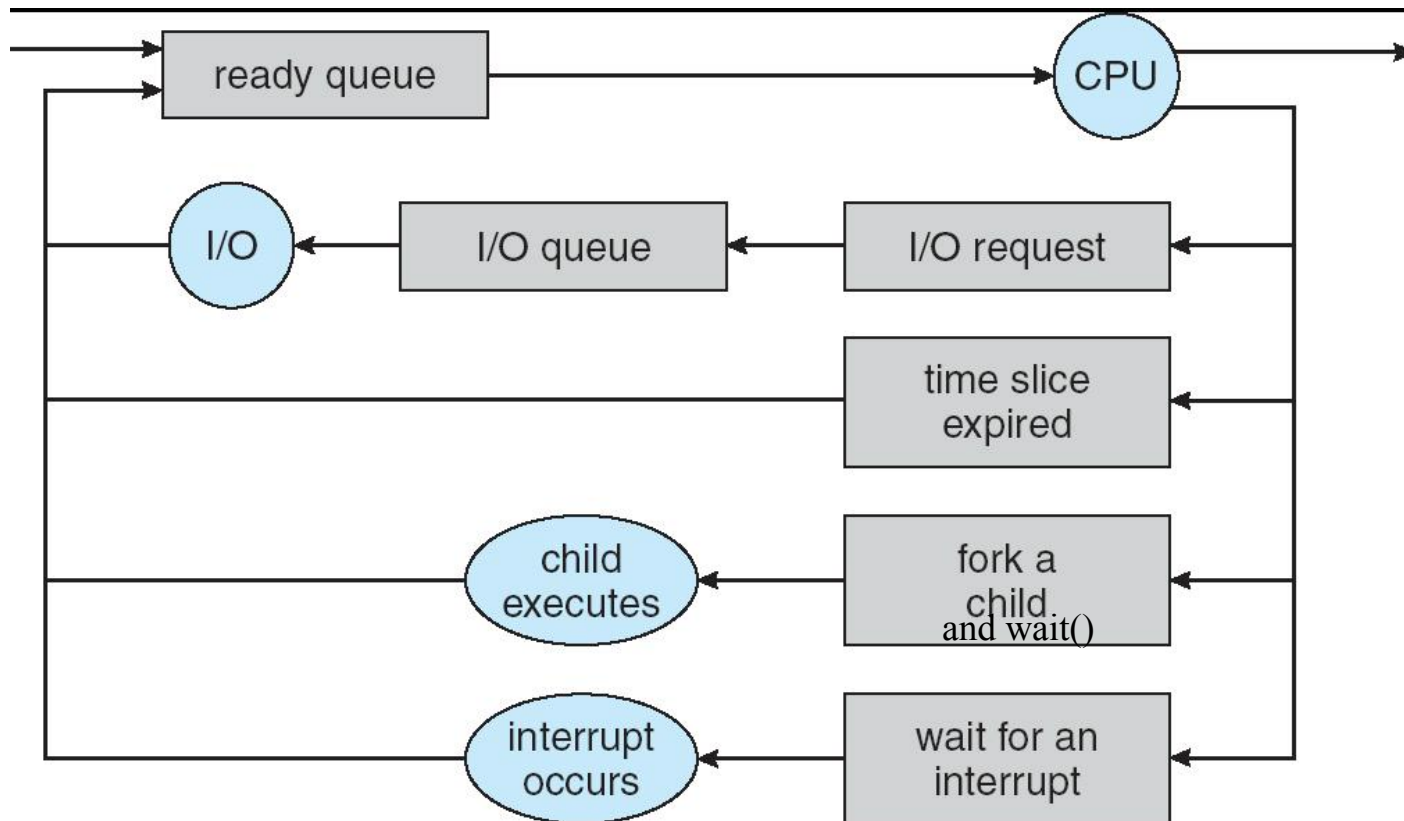
when more than two processes

Process scheduling

- Multiprogramming
 - utilization
- Timesharing
 - interactive
- Scheduling queues
 - linked list
 - ready queue
 - I/O queue



Queuing system



Queuing scheduler

- Who's the next?
- Long-term scheduler
 - job scheduler (spooling)
 - get to the ready queue
 - CPU-intensive vs I/O intensive
- Short-term scheduler
 - CPU scheduler
 - frequency vs overhead

More on scheduling

- Medium-term scheduler
 - who is NOT the next
 - reduce the degree of multiprogramming
 - swap-in/out
- Scheduling algorithms
 - first-come-first-server, shortest-job-first, priority, round-robin, fair and weighted fair, ...
 - more in Chapter 5

This lecture so far

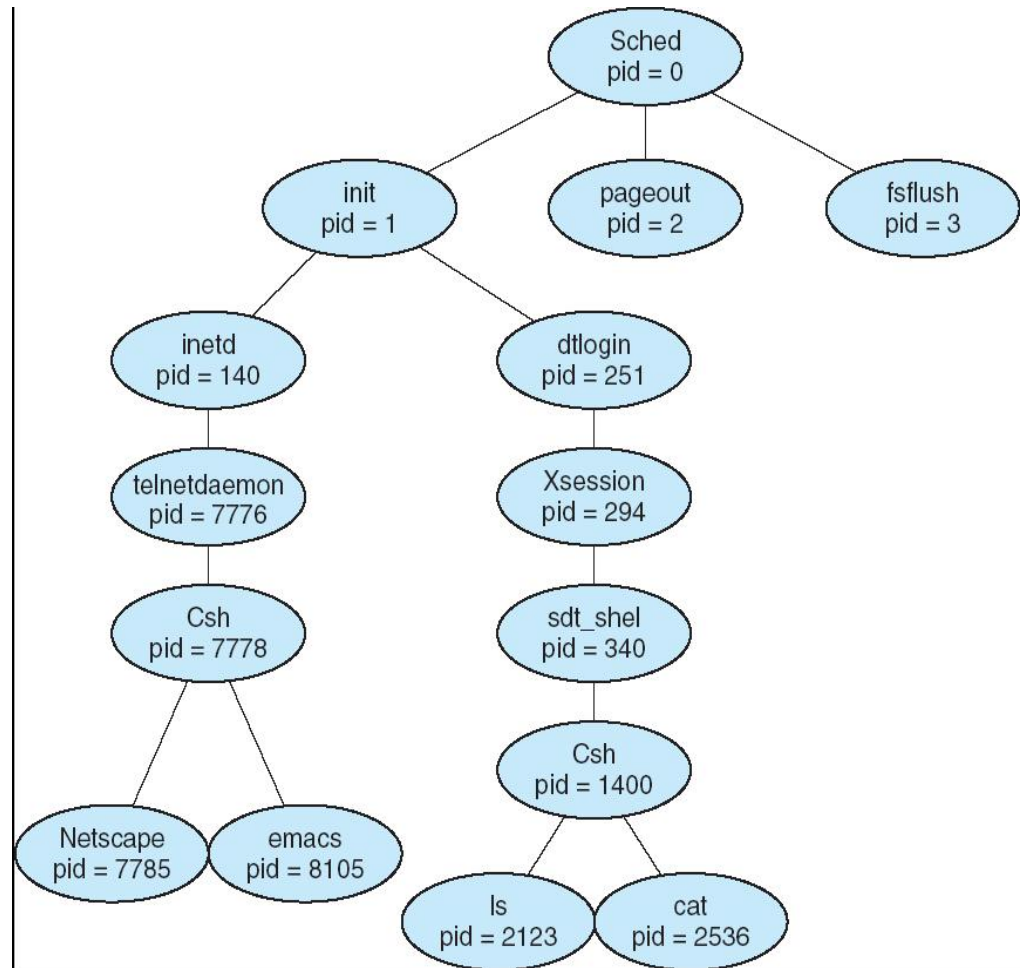
- Process and process scheduling
 - process vs program
 - process control block
 - context switch: what to save/restore
 - process scheduling
- Explore further
 - process status: `/bin/ps`
 - top CPU processes: `/usr/bin/top`

Process creation

- Creating processes
 - parent process: create child processes
 - child process: created by its parent process
- Process tree
 - recursive parent-child relationship; why tree?
 - `/usr/bin/pstree`
- Process ID (PID) and Parent PID (PPID)
 - usually nonnegative integer

Process tree

- sched (0)
 - init (1)
 - all user processes
 - pageout
 - memory
 - fsflush
 - file system



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pstree on linux.csc.uvic.ca

```

snmpd
sshd
  5*[sshd—sshd—sftp-server]
  sshd—sshd—bash—ssi
  sshd—sshd—bash—more
  sshd—sshd—bash—pstree
  sshd—sshd—csh—sftp-server
  sshd—sshd—bash
  sshd—sshd—bash—mutt
  sshd—sshd—tcsh—nano
  sshd—sshd—tcsh
syslogd
  
```

Parent vs child processes

- Process: running program + resources
- Resource sharing: possible approaches
 - all shared, or
 - some shared (e.g., read-only code), or
 - nothing shared*
- Process execution: possible approaches
 - parent waits until child finishes, or
 - parent and child run **concurrently***

fork(), exec*(), wait()

- Create a child process: fork()
 - return code < 0: error (in “parent” process)
 - return code = 0: you’re in child process
 - return code > 0: you’re in parent process
 - return code = child’s PID
- Child process: load a new program
 - exec*(): front-end for execve(file, arg, environ)
- Parent process: wait() and waitpid()

exec*()

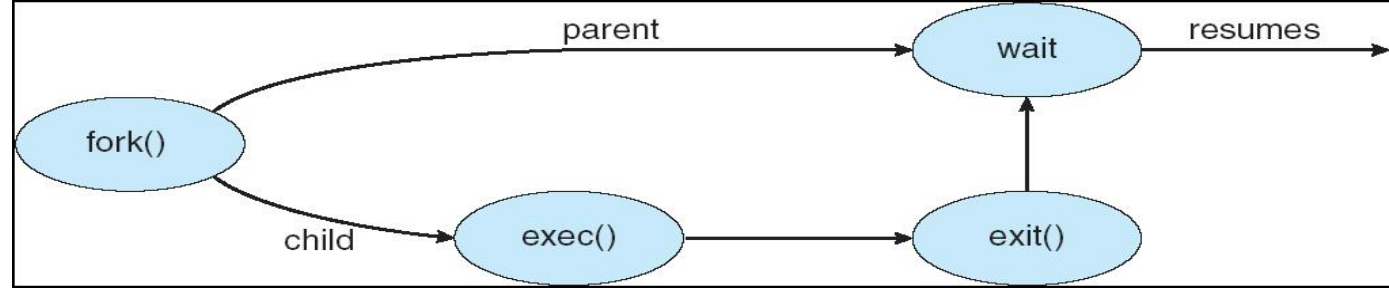
- run the command `ls -l /usr`

```
// execl (List, No Path Lookup)  
execl("/bin/ls", "ls", "-l", "/usr", NULL);
```

```
// execlp (List + Path Lookup)  
execlp("ls", "ls", "-l", "/usr", NULL);
```

```
// execv (Vector, No Path Lookup)  
char *args[] = {"ls", "-l", "/usr", NULL}  
execv("/bin/ls", args);
```

```
//execvp (Vector + Path Lookup)  
char *args[] = {"ls", "-l", "/usr", NULL};  
execvp("ls", args);
```



Example

```
int main()
{
    Pid_t pid;
    /* fork another process */
    pid = fork();
    if (pid < 0) { /* error occurred */
        fprintf(stderr, "Fork Failed");
        exit(-1);
    }
    else if (pid == 0) { /* child process */
        execvp("/bin/ls", "ls", NULL);
    }
    else { /* parent process */
        /*parent will wait for the child to complete*/
        wait (NULL);
        printf ("Child Complete");
        exit(0);
    }
}
```

CS6360
* what if no wait()?

Win In-class Bonus (1 point)

- Submit your answer within 10min from the start
 - for the following two questions, 0.4 points each
 - 0.2 points for active participation (as long as you submit, name, V#, A0#)
 - group discussion is allowed
 - no internet searching
- How to use your bonus grades
 - $\text{Final} = \min\{100, \text{weighted grades}(\text{exams, assignments}) + \text{bonus}\}$
- More bonuses on the way

Here is the code of file `ex1.c`:

Q1

```
# include <stdio.h>
# include <sys/types.h>
# include <unistd.h>

int main()
{
    fork();
    fork();
    fork();
    printf("Hello CSC360! PID = %d\n", getpid());
    return 0;
}
```

After executing the commands `gcc ex1.c` and `./a.out`, how many lines of `"Hello CSC360! PID = ###"` we will have?

Here is the code of file `ex1.c`:

```
# include <stdio.h>
# include <unistd.h>
# include <stdlib.h>

int main(int argc, char *argv[])
{
    printf("PID of ex1.c = %d\n", getpid());
    char *args[] = {"Hello", "CSC", "360", NULL};
    execv("./ex2", args);
    printf("Back to ex1.c");
    return 0;
}
```

And here is the code of file `ex2.c`:

```
# include <stdio.h>
# include <unistd.h>
# include <stdlib.h>

int main(int argc, char *argv[])
{
    printf("We are in ex2.c\n");
    printf("PID of ex2.c = %d\n", getpid());
    return 0;
}
```

After executing the commands `gcc ex1.c -o ex1`, `gcc ex2.c -o ex2`, and `./ex1`, the output will be:

Q2

- A. PID of ex1.c = 5962
We are in ex2.c
PID of ex2.c = 5962
- B. PID of ex1.c = 5962
We are in ex2.c
PID of ex2.c = 5963
- C. PID of ex1.c = 5962
Back to ex1.c
- D. PID of ex1.c = 5962
We are in ex2.c
PID of ex2.c = 5962
Back to ex1.c

Process termination

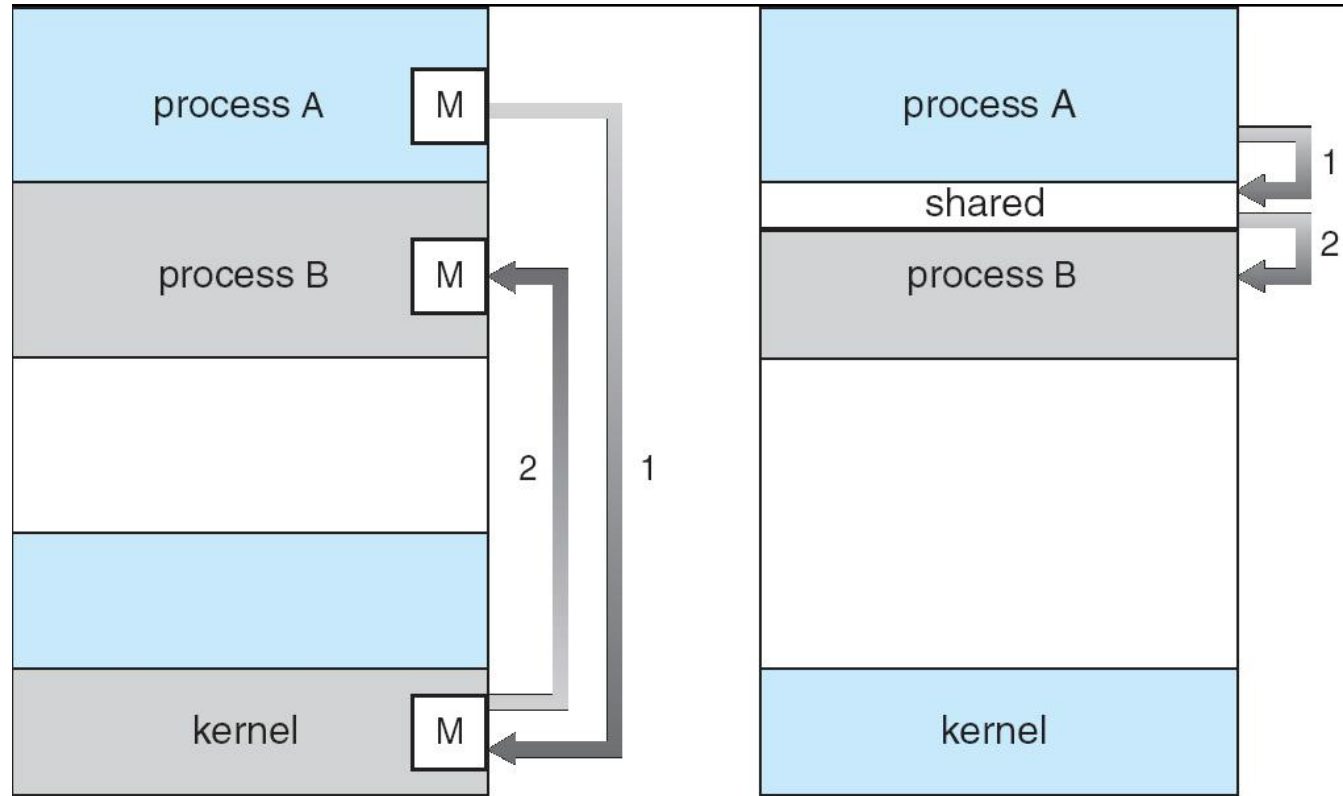
- Terminate itself: `exit()`
 - report status to parent process
 - release allocated resources
- Terminate child processes: `kill(pid, signal)`
 - actually send a signal to the child
 - child resource exceeded, child process no longer needed, and so on
 - parent is exiting
 - cascading termination, or find another parent

Process communication

- Independent process
 - standalone process
- Cooperating process
 - affected by or affecting other processes
 - sharing, parallel, modularity, convenience
- Process communication
 - shared memory
 - message passing

Message passing vs shared memory

- Overhead
- Protection



(a)

CSc 360

(b)

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* again pros and cons

The 2nd half of this lecture

- Process operations
 - process creation
 - process tree
 - process termination
 - the need for inter-process communication
- Explore further
 - /bin/ps, /usr/bin/top, /usr/bin/pstree
 - how does a child process find its parent's PID?

Next lecture

- Inter-process communication
 - read OSC7 Chapter 3 (or OSC6 Chapter 4)